

YOLO26n-sem Performance & Inference Report

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1. System Configuration & Architecture Specs

Model Name	yolo26n-sem.pt
Input Dimensions	2048x1024 (WxH)
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/PT_Output
OS	Ubuntu 24.04.4 LTS
Python Version	3.12.3
PyTorch Version	2.13.0+cu130
Ultralytics Version	8.4.75
Inference Device	NVIDIA RTX A4000
CUDA Version	13.0
Model Architecture	YOLO-semantic
Total Parameters	1,627,822
File Size	3.33 MB
Number Of Classes	19
Class Names	road, sidewalk, building, wall, fence, pole, traffic light, traffic sign, vegetation, terrain, sky, person, rider, car, truck, bus, train, motorcycle, bicycle
Precision Mode	FP32

2. Pipeline Speed & Throughput Performance

Total Images Processed	5
Inference Time (time.perf_counter)	708.01 ms
Inference Time (YOLO Results)	228.31 ms
Throughput	7.06 Images/sec
Peak Memory Usage	352.35 MB

3. Image-Level Class Mask Breakdown

Target File Context	Segmented Class Name	Pixel Spatial Area	Total Area Coverage
Cat_November_2010-1a.jpg.webp	road	77,042 px	6.26%
	building	545,121 px	44.29%
	wall	77 px	0.01%
	vegetation	462,725 px	37.60%
	terrain	799 px	0.06%
	sky	128,486 px	10.44%
	person	13 px	0.00%
	car	16,457 px	1.34%
images.jpeg	road	7,269 px	2.37%
	sidewalk	96,653 px	31.49%
	building	12,832 px	4.18%
	wall	59 px	0.02%
	pole	382 px	0.12%
	vegetation	14 px	0.00%
	terrain	1,970 px	0.64%
	sky	139,592 px	45.48%
	person	48,008 px	15.64%
	car	136 px	0.04%
images1.jpeg	road	23,737 px	7.74%
	sidewalk	3,542 px	1.16%
	building	193,322 px	63.06%
	wall	5,946 px	1.94%
	fence	1,750 px	0.57%
	pole	43 px	0.01%
	vegetation	49,758 px	16.23%
	terrain	14,130 px	4.61%
	person	13,499 px	4.40%
	car	803 px	0.26%

Target File Context	Segmented Class Name	Pixel Spatial Area	Total Area Coverage
istockphoto-1189925691-612x612.jpg	bicycle	30 px	0.01%
	road	42,448 px	17.00%
	sidewalk	6,383 px	2.56%
	building	3,143 px	1.26%
	fence	163 px	0.07%
	pole	766 px	0.31%
	traffic sign	378 px	0.15%
	vegetation	96,287 px	38.56%
	terrain	792 px	0.32%
	sky	1,600 px	0.64%
tigermain.jpg	car	411 px	0.16%
	bus	97,325 px	38.98%
	road	110,538 px	4.50%
	sidewalk	4,134 px	0.17%
	building	333,730 px	13.58%
	wall	473,619 px	19.27%
	fence	276 px	0.01%
	pole	116 px	0.00%
	vegetation	706,058 px	28.73%
	terrain	513,060 px	20.88%
	person	247,736 px	10.08%
	car	68,183 px	2.77%
	bicycle	150 px	0.01%